5 CMOS Testing

CMOS Testing

$$\text{Yield} = \frac{\text{No. of good die}}{\text{Total No. of die in a wafer}}$$

Testing is required at all levels for a good yield.

Testing a die (chip) can occur:

- at the wafer level
- at the packaged chip level
- at the board level
- at the system level

1. By finding a defected chip at an earlier level, the manufacturing cost may be reduced.
2. A component vendor can only test at the wafer or chip level special systems, such as satellite-borne electronics.
3. The test may fall into two main categories.
   a. Functionality tests verify the chip functioning. They are used early in design cycle.
   b. Manufacturing tests verify that every gate and register function correctly. They are used after the chip is manufactured.

Functionality Test

1. Does this adder add?
2. Does this counter count?
3. Does this state machine yield the right outputs at the right click cycles?

At each level of the design hierarchy such small tests are written to verify the functional equivalence between the new higher and lower level.

Manufacturing Tests

1. Manufacturing defects may occur during fabrication.
   a. Layer-to-layer shorts
   b. Discontinuous wires
   c. Thin oxide shorts to substrate or well

   These may lead to:
   a. nodes shorted to ground or power lines
   b. nodes shorted to each other
   c. inputs or outputs disconnected
2. Functionality tests verify the function of a chip as a whole, whereas manufacturing tests verify that every gate operates as expected. Tests are normally carried out at the wafer level to cell out bad die, and then, on the packaged parts.
3. Apart from the verification of internal gates, I/O integrity is also tested through I/O level test, speed test, and I_{DD} test.

Fault Models

Stock at O (SA0)
Stock at 1 (SA1)

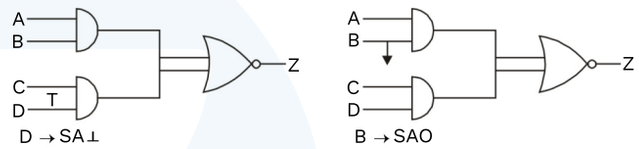


Fig. 5.1

Such faults occur due to thin oxide shorts or metal to metal shorts.

Short Circuit and Open Circuit Faults

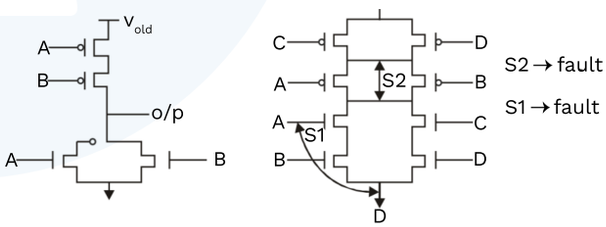


Fig. 5.2

Such faults occur due to missing gate, source, or drain connections, or if source and drain diffusions shorted.

Observability: the observability of a particular internal circuit node is the degree to which one can observe the node at the output of an IC.

This measure is of importance when a designer/ tester desires to measure the output of a gate within a larger circuit to check that it operates correctly.

It should be the aim of well-designed chip to have easily observed gate output.

Controllability: the controllability of an internal circuit node within a chip is the measure of the case of setting the node to a 1 or 0 state. This measure is of importance when assessing the remove difficulty of testing a particular signal within a circuit. It should be the aim of a well-designed circuit to have all nodes easily controllable.

Automatic Test Pattern Generation (ATPG)

ATPG is a method to generate test vectors. Here is an example.

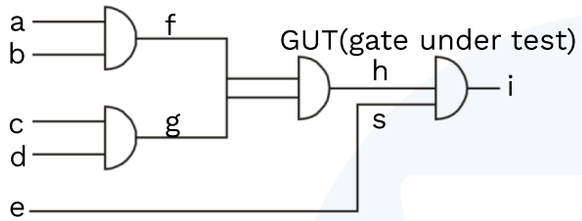


Fig. 5.3

If we are checking for on SA 1 fault at node h, we must be able to set it to 0. Test vector would be (for SA1):

{a, b, c, d, e} = {0, 1, x, x, 1} or {1, 0, x, x, 1} or {0, 0, x, x, 1} or {1, 1, 0, 0, 1}.

For SAO:

{a, b, c, d, e} = {1, 1, 0, 1, 1} or {1, 1, 1, 0, 1}

Design Strategies for Test

Design for testability: it means designing a circuit that is testable.

Controllability: it is the ability to set (1) and reset (0) every node internal to the circuit.

Observability: it means to observe either directly or indirectly the state of any node in the circuit.

Three Approaches to DFT

1. Ad hoc testing
2. Scan-based approaches
3. Self-test and built-in testing

Ad hoc Testing

One way to increase the testability is to make nodes more accessible at some cost by physically inserting more access circuits to the original design. Following are some of the ad hoc testable design techniques.

- Partition and mux technique
- Disable internal oscillators and clocks
- Avoid asynchronous logic and redundant logic
- Avoid delay dependent logic

Partition and Mux Technique

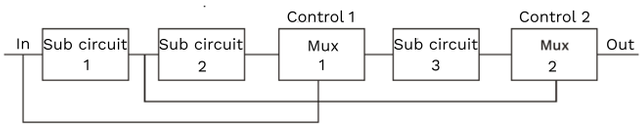


Fig. 5.4 Partition and Mux Method for Large Circuits

Large circuits can be partitioned and muxes can be inserted so that some of the primary inputs can be fed to partitioned parts through muxes accessible control signals. For example: 32 bit counter can be divided into two 16 bit counters. This technique may increase the chip area and circuit delay. It is similar to divide and conquer approach.

Disable Internal Oscillators and Clocks

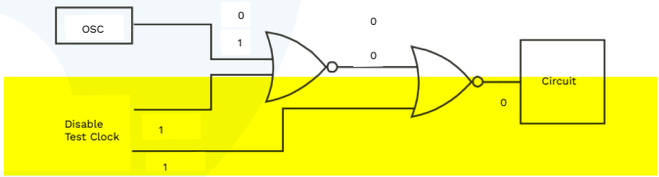


Fig. 5.5

To avoid synchronisation problem during testing, internal oscillators and clocks should be disabled.

Avoid Asynchronous and Redundant Logic

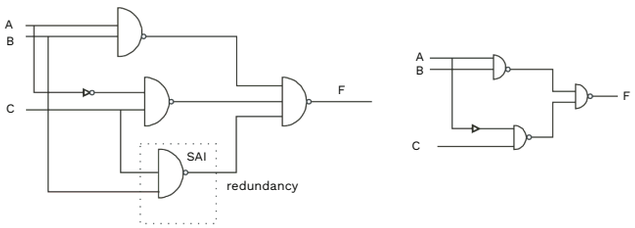


Fig. 5.6

$F = AB + BC + \bar{A}C$

$= AB + \bar{A}C$

The design and test of asynchronous logic circuits are more difficult than for synchronous logic circuits.

Operation of an asynchronous logic circuit is sensitive to input patterns, often causing race problems and hazards of having momentary signal values opposite to the expected values.

SA $\perp$ fault on its output line cannot be detected.

So, this gate can be removed while testing.

Use of redundant circuit can make test generation complex and difficult.

Avoid Delay Dependent Logic

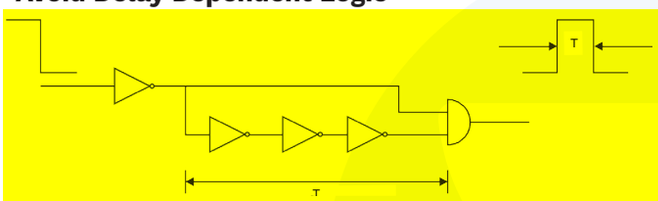


Fig. 5.7 A Pulse Generation Circuit Using a Delay Chain of Three Inverters

Such delay dependant logic is viewed as redundant combinational logic, and thus, lead to avoid in design for testability.

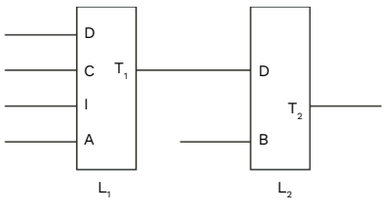
Scan-Based Technique

Structured approach to testability.

Level Sensitive Scan Design (LSSD)

There are two requirements.

1. Circuit level sensitive
2. Each register connected to a serial shift register



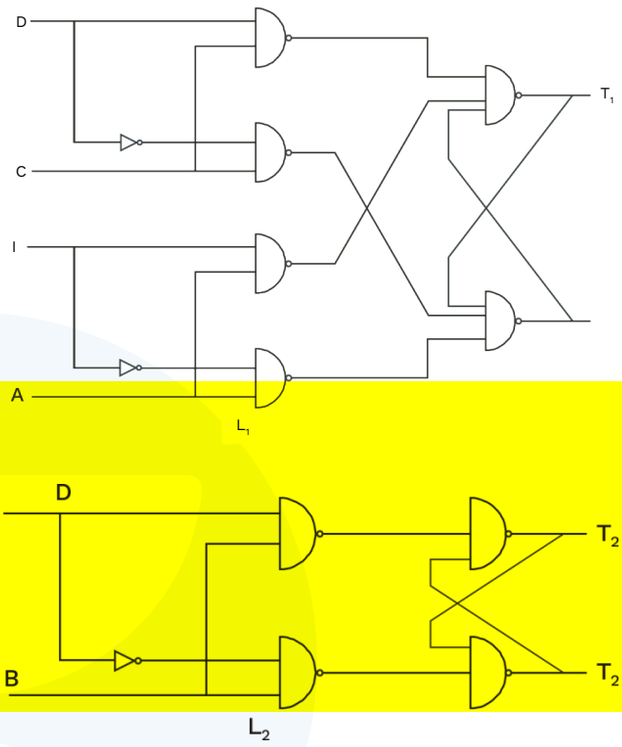


Fig. 5.8 A Shift Register Latch

C, A: Enable signals

D, I: Data ports

A and C: cannot be simultaneously high

L_1: Master

L_2: Slave

- When A is high, L_1 (T_1) is set by I
- When C is high, L_1 (T_1) is set by D
- When B is high, T_1 is passed to T_2

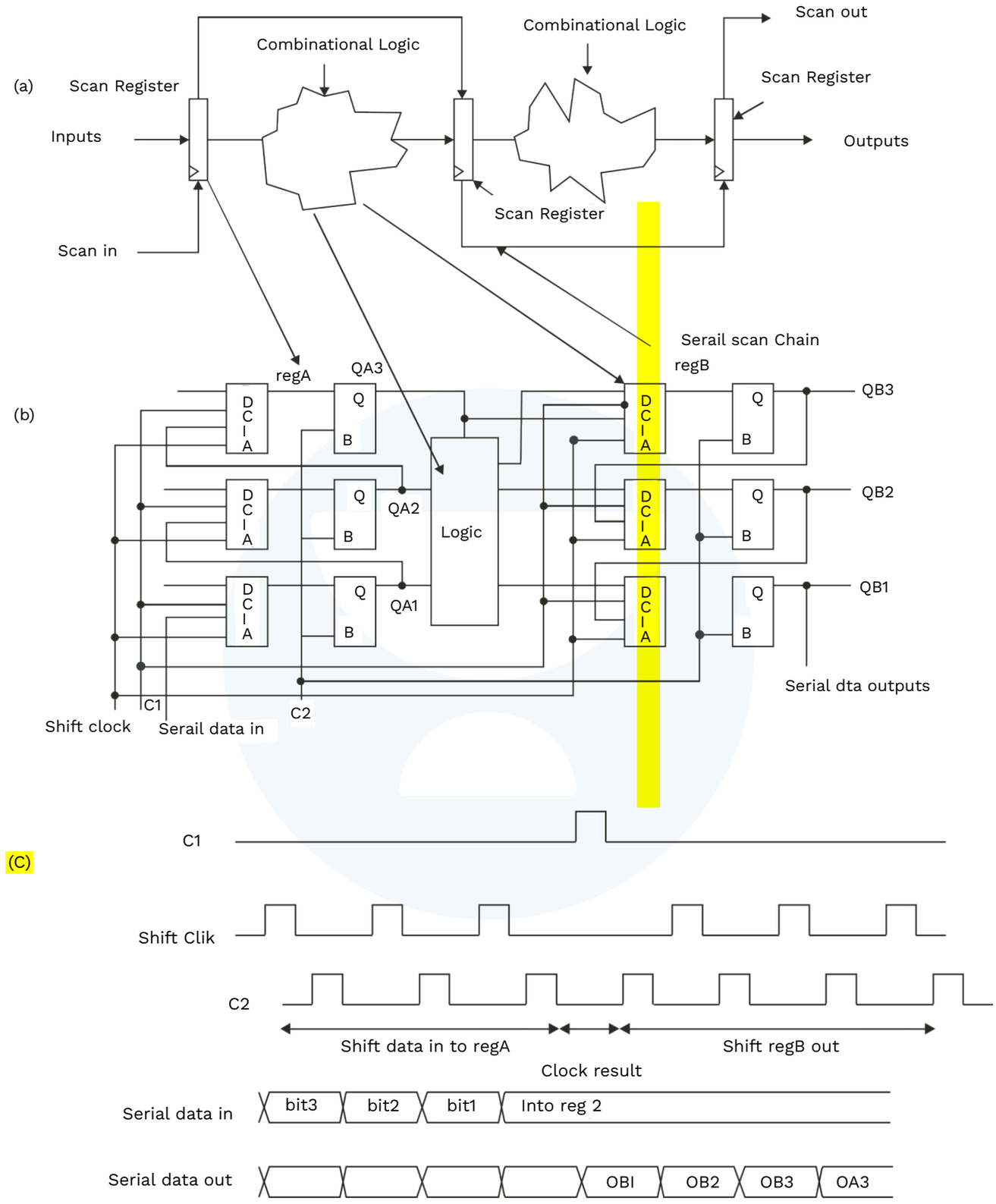


Fig. 5.9 An LSSD Scan Chain: (a) Basic Architecture, (b) Example Circuit, and (c) Example Timing

The prime disadvantage is the complexity of shift register latch (SRL) impacting density and speed.

Self Test Techniques

Signature Analysis and BILBO

It involves the use of a pseudo random sequence generator (PRSG) to generate input signals for a section of combinational circuit, and then, using a signature analyser to observe the opposite signals.

The PRSG implements a polynomial of some length. It is constructed from a Linear Feedback Shift Register (LFSR), which is constructed in turn from a number of 1-bit shift registers connected in a serial fashion as shown in figure given below. Signature analysis can be merged with the scan technique to create a structure known as built in logic observation (BILBO).

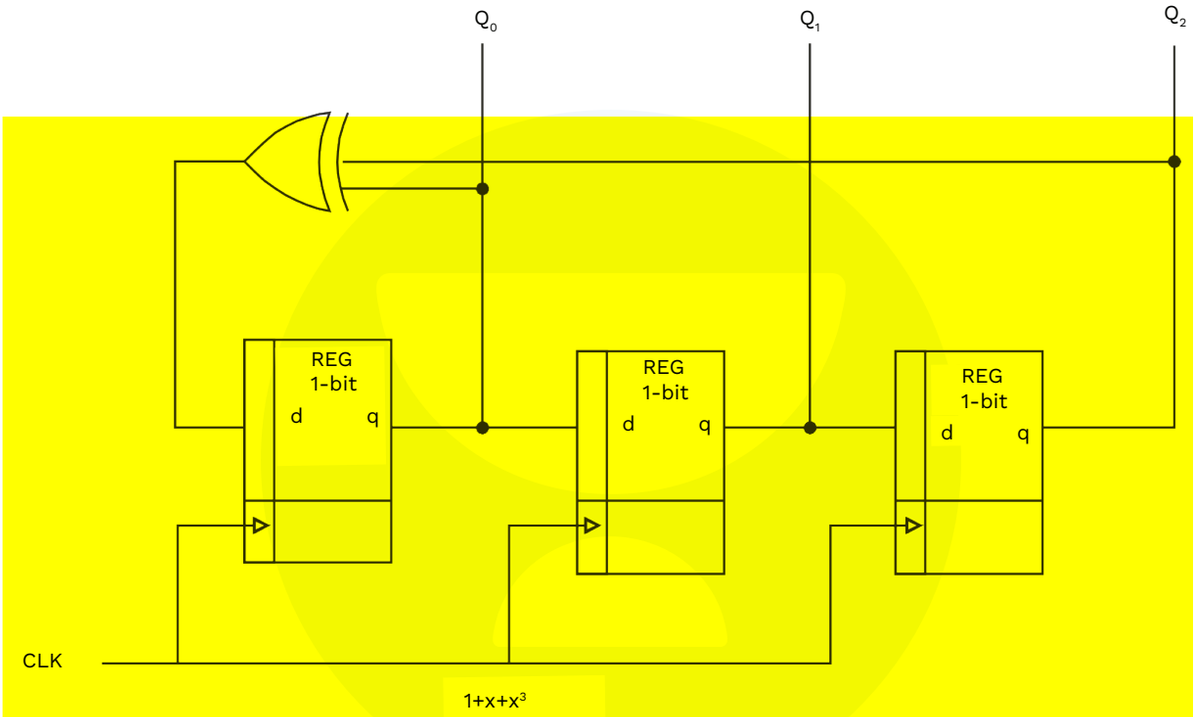
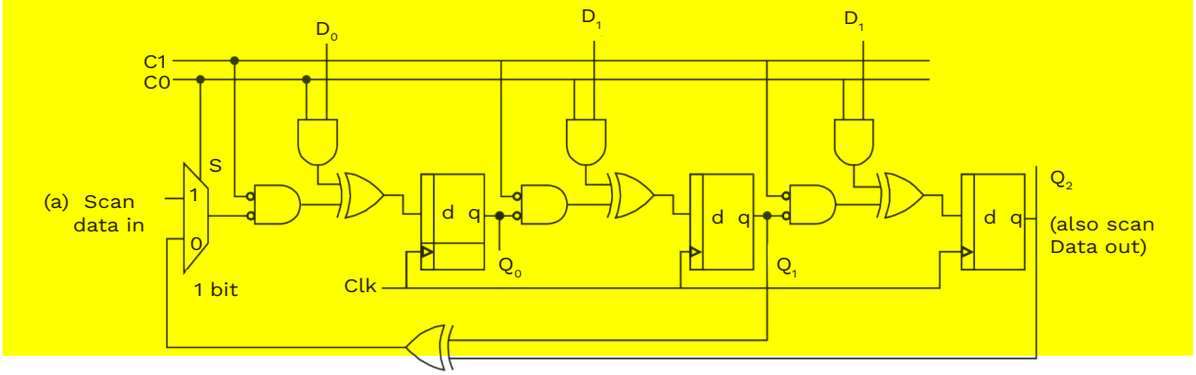


Fig. 5.10 Pseudo Random Sequence Generator



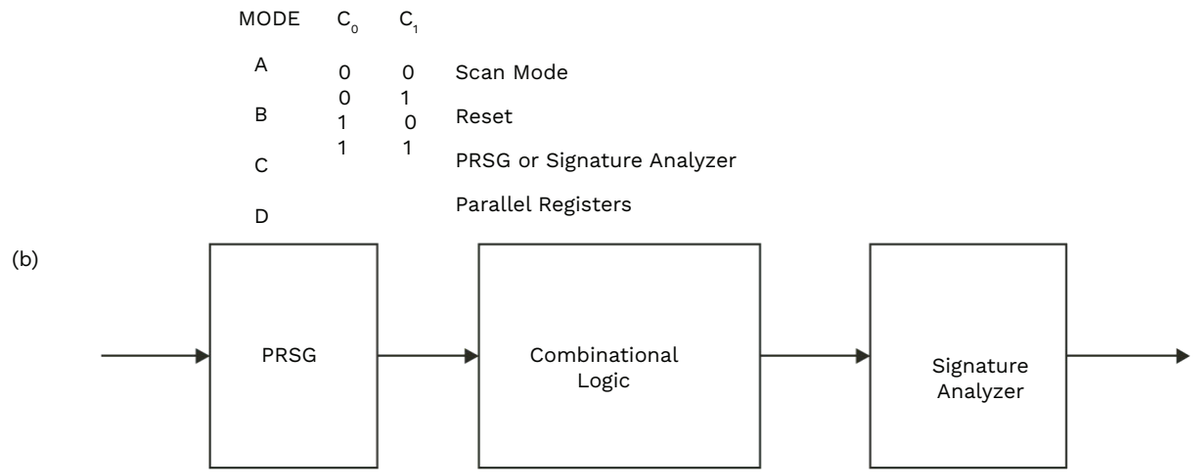


Fig. 5.11 Built in Logic Block Observation (BILBO): (a) Individual Register and (b) Use in a System

Practice Exercise

Level I

1. Testing of chip is done:
(A) at the wafer level
(B) at the packaged chip level
(C) at the board level
(D) All of the above

2. What happens by identifying a defected chip at an earlier level?
(A) Manufacturing cost may be reduced
(B) Time and energy invested can be saved
(C) Does not help
(D) (A) and (B) only

3. The functions performed during chip testing are:
(A) detect faults in fabrication
(B) detect faults in design
(C) finding failures in functionality
(D) All of the above

4. ATPG stands for:
(A) Attenuated Transverse Wave Pattern Generation
(B) Automatic Test Pattern Generation
(C) Aligned Test Parity Generator
(D) None of the above

5. Approach to DFT is:
(A) Ad hoc testing
(B) scan-based approach
(C) self-test and built-in test
(D) All of the above

6. A metallic blob present between drain and ground acts as:
(A) physical defect
(B) logical fault
(C) electrical fault
(D) All of the above

7. The ease with which the controller establishes specific signal values at each node by setting input values is known as:
(A) testability
(B) controllability
(C) observability
(D) manufacturability

Level II

1. Divide and conquer approach to the large and complex circuits for testing is found in:
(A) partition and mux technique
(B) ATPG method

(C) scan-based technique
(D) All of the above

2. LSSD stands for:
(A) Level Sensitive Scan Design
(B) Level Sensitive System Design
(C) Level Sensitive Scan Detection
(D) Linear System Scan Design

3. Internal oscillators and clocks are disabled during testing:
(A) to avoid synchronisation problems
(B) to increase testing speed
(C) to avoid delay problems
(D) None of the above

4. Scan-based technique is a/an:
(A) easy approach to testability
(B) structured approach to testability
(C) difficult approach to testability
(D) only approach to testability

5. PRSG stand for:
(A) Polynomial Random Scale Generator
(B) Pseudo Random Sequence Generator
(C) Pseudo Random Scale Generator
(D) Pseudo Random Scan Generator

6. BILBO stands for:
(A) Built-in Logic Block Observation
(B) Big Input Line Block Observation
(C) Built-in Logic for Block Output
(D) Big Input Length Big Output

Answer Key

Level I

1. (D) **2.** (D) **3.** (D) **4.** (B) **5.** (D) **6.** (D) **7.** (B)

Level II

1. (A) **2.** (A) **3.** (A) **4.** (B) **5.** (B) **6.** (A)